Modular Acoustic Suppression System (MASS)

Emmi Cayer, Gabriel Choi, Jonathan Corvera, Lucas Kaemmerer, Julia Megargle, Michael Milner

Advisor: Rick Lasko

Sponsor: Dr. John Chen

Background

Currently, the Environmental Control and Life Support Systems (ECLSS) in the human lander produces noise outside of the specified Noise Criterion levels affecting astronauts' sleep, concentration, and communication.

Objective: Design an innovative solution to reduce noise generated by ECLSS systems in long-duration space habitats.

Design Specifications

System Constraints:

- Minimal barriers to NASA adoption (low mass, low power consumption)
- No additional risks to crew
- Mission operational life of 30-days for lunar missions, 1200-days for Mars missions

Design Specifications:

Specification	Required Value	Design Value
Acoustic	63-250 Hz, 50-64 dB	---
Weight	<100 lb	4.33lbs
Power	<0.95 kW	0.136 kW
Volume	<1.5 cubic feet	< 1.0 cubic feet

Manufacturing and Assembly

3D Printing Operations:

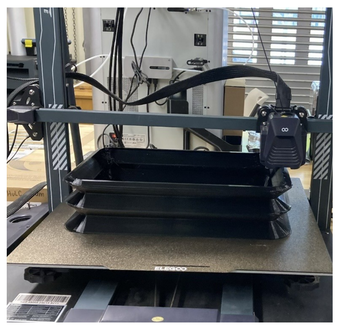
- Reverberation chamber and gasket - TPU filament
- Top/bottom mounting plates, linear rail mounts - PLA filament

Lathe Operations:

- Power screw turned down to size

Waterjet Operations:

- Top mounting plate support

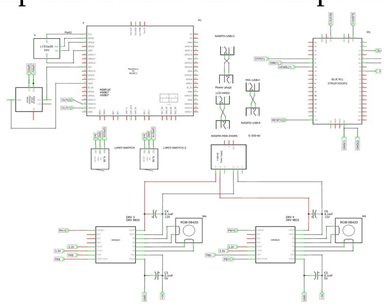


3D Printing Fabrication

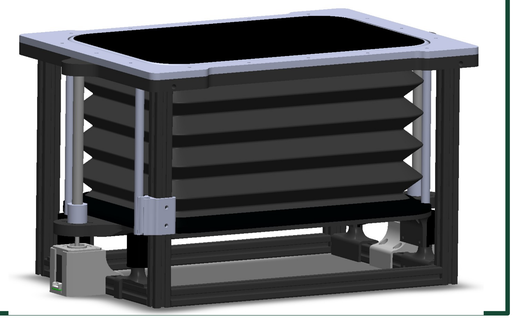
Final Design

Key Features:

- Noise Reduction – Flexible membrane reduces sound by achieving resonance with incoming sound waves.
- Adjustable – Chamber height is adjusted by linear actuators that respond to varying frequency inputs, changing membrane's resonant frequency.
- Compact – Components contained in 18in x 12in x 12in box, making implementation and transportation easy.



Electrical System Design

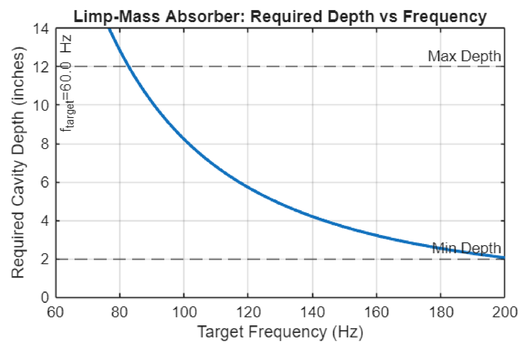


Mechanical System Design

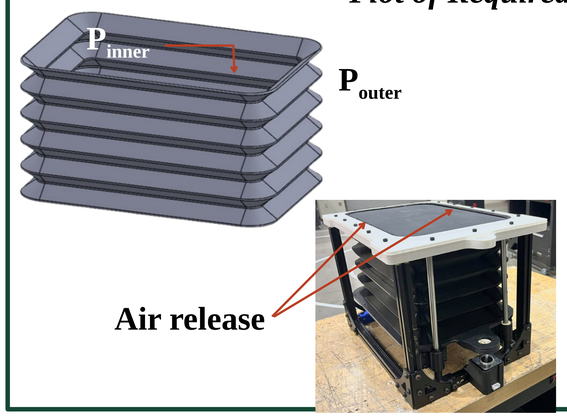
Analysis

Membrane Selection:

- Analyzed as oscillating mass-acoustic spring-damper
- Required depth plotted and analyzed in MATLAB



Plot of Required Depth for Target Frequencies



Pressure Equalization:

- Functions by equalizing to surrounding pressure during operation ($P_{inner} = P_{outer}$)
- Air released through small gaps between membrane and gasket allowing equalization while maintaining constant air volume during operation

Testing and Results

Test 1: Linear Actuator and Electronic Verification

- Sweep of frequencies ranging from 60-250 Hz
- Ensured actuators adjusted to changing frequencies, switching directions and height (6" max) dependent on frequency spectrum
- Smoothed frequency enables consistent transition despite highly variable sound conditions

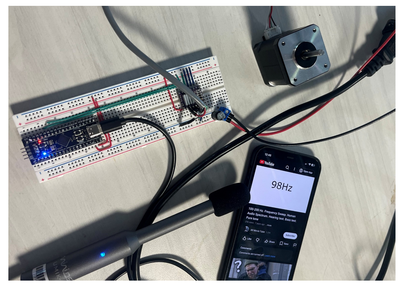
Detected Frequency [Hz]	Smoothed Frequency [Hz]	Required Depth [in]	Confidence (>5 is good)
128.91	131.04	4.80	11.33
175.78	139.28	4.26	6.57
234.34	188.32	2.32	5.55

Test 2: Pressure Equalization Verification

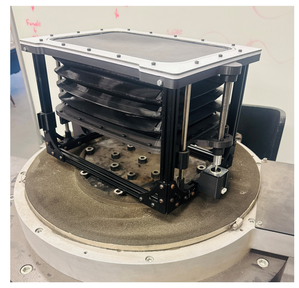
- Chamber depth shifted 3 inches to verify pressure equalization at varying depths throughout 5 tests
- Successfully equalized after 4.49 seconds on average

Test 3: Vibration Test

- Verified mechanical system's structural integrity at critical resonant frequencies



Stepper Motor Testing Setup



Vibration Testing Setup

Conclusion and Recommendations

- Prototype successfully verifies acoustic control
- system and pressure equalization
- Recommend material switch to silicone for weight reduction and easier height adjustment as well as extending frequency range
- Further testing required in acoustic chamber to verify noise suppression